

Technical, economic, and environmental evaluation of a grid-connected hybrid PV/biogas generator/battery microgrid system

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ABSTRACT

This study presents a techno-economic and environmental analysis of a grid-connected hybrid microgrid designed for a university campus in Douala, Cameroon, using HOMER Pro software. The proposed system, comprising a 107-kW photovoltaic (PV) array, a 500-kW biogas generator fed by human biomass, 8 battery units, and a 93-kW backup diesel generator, is designed to meet a daily load of 580.60 kWh/d. This hybrid configuration is compared to a conventional system (Grid/Diesel). The results show that the hybrid system is significantly more cost-effective, with a Levelized Cost of Energy (LCOE) of \$0.07563/kWh compared to \$0.1597/kWh for the conventional model. The Net Present Cost (NPC) is reduced from \$437,605.20 to \$300,661, with an initial investment of \$84,603 and a payback period of 4.96 years. Environmentally, the hybrid system reduces CO₂ emissions by nearly 50 % by leveraging renewable sources, which contribute to 40 % (PV) and 25.3 % (biogas) of the total energy production. This study demonstrates the viability of using local biomass resources to enhance energy autonomy, reduce operational costs, and improve the environmental footprint of institutional facilities.

Abbreviations	Signification
BEAC	Bank of Central African States
CAMES	African and Malagasy Council for Higher Education
CC	Cycle Charging
COE	Cost of Energy
COP21	21st Conference of the Parties (on climate change)
CRF	Capital Recovery Factor
ESIA	Environmental and Social Impact Assessment
HOMER	Hybrid Optimization of Multiple Energy Resources
HRES	Hybrid Renewable Energy Systems
ICE	Internal Combustion Engine
IRR	Internal Rate of Return
IUT	University Institute of Technology
LCOE	Levelized Cost of Energy
LF	Load Following
LHV	Lower Heating Value
MG	Microgrid
MINEPEDD	Ministry of Environment, Nature Protection and Sustainable Development
NPC	Net Present Cost
NPV	Net Present Value

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Abbreviations	Signification
O&M	Operation and Maintenance
PSO	Particle Swarm Optimization
PV	Photovoltaic
RE	Renewable Energy
ROI	Return on Investment
SAM	System Advisor Model
SOC	State of Charge
STC	Standard Test Conditions
TS	Total Solids
UIT	University Institute of Technology
VS	Volatile Solids
WHO	World Health Organization
WOA	Whale Optimization Algorithm
Symbol	Meaning
A _g	Fuel consumption coefficient constant
B _g	Fuel consumption coefficient constant
B _c	Battery capacity
C _{ann,tot}	Total annualized cost of the system

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Abbreviations	Signification
C_{bat}	Battery bank's capacity
C_{CO_2}	Environmental damage cost associated with CO ₂ emissions
$CRF(i, R_{proj})$	Capital Recovery Factor
E_i	Total energy needed for powering the storage device
E_{gen}	Sum of the energy produced by the microgrid and the grid
$E_{gen, MG}$	Energy produced by the microgrid
E_{grid}	Total energy purchased from the grid annually
f	Nominal inflation rate
f_{PV}	PV derating factor
F_{CO_2}	CO ₂ emission factor of the power generation system
F_{dg}	Fuel consumption of a diesel generator
G_T	Solar radiation incident on the PV generator at the current time step
$G_{T, STC}$	Radiation under standard test conditions
i	Real annual interest rate (net)
i'	Nominal interest rate
$I_{bat}(t)$	Battery current at time t
ϕ_{CO_2}	Social cost of carbon
P_{dg}	DG rated power
P_{dg-out}	DG output power
P_{PV}	Power output of a PV solar panel
Q	Battery's electrical charge
R_{proj}	Project duration
SOC_i	Vehicle battery's state of charge
SOC_t	Battery's immediate state of charge at time t
SOC_{t-1}	Battery's self-discharge rate at t-1 hours
T_{amb}	Ambient temperature
T_C	PV cell temperature
$T_{C, STC}$	PV cell temperature under standard test conditions
V	Battery's voltage
y_{PV}	PV generator's rated output under standard test conditions
α_p	Temperature coefficient of power
η_{bat}	Battery charge efficiency
$\eta_{charger}$	Battery charger's efficiency
σ	Battery's self-discharge rate (implied in formula)
τ	Plant's life span (same as R_{proj})

1. Introduction

1.1. Background and motivation of the study

Renewable energies such as wind, biomass, solar power, and hydropower are naturally regenerated sources of power and are viewed as critical components of the following day's energy system, which strives to be more energized decentralized, and sustainable. Driven by concerns about climate change and global warming, the global installed capacity of renewable energy increased by 50 % in 2023. By the end of that year, the total global renewable energy installations including solar, wind, hydropower, geothermal, marine, and biomass reached approximately 3,372 GW, with 1,600 GW from solar photovoltaic systems and 151 GW from biomass energy. This growth in the renewable energy market reflects a global shift toward more sustainable and environmentally friendly energy technologies [1].

Since the 2015 United Nations Climate Conference (COP21), over 164 countries have pledged to increase their use of renewable energy [2]. For instance, the European Union wants to have at least 32 % renewables in its overall energy supply by 2030, while China plans to generate more than 20 % of its power from such resources by 2030 [3]. By 2050, renewable energy sources are predicted to generate the majority of the nation's power, and the United States is gradually increasing its installed renewable energy capacity even in the lack of government objectives [4].

Renewable energy is crucial to addressing Africa's ongoing energy problems, particularly in Cameroon, which has a high reliance on fossil fuels, frequent power outages, and poor energy management [5]. Despite having a lot of potential for renewable energy, Cameroon has a lot of natural resources and a climate that is conducive to solar energy, but few of these have been integrated into the country's power system. The main issues are variations in energy generation, the intermittent

nature of renewable sources, and the low density of energy of one source of energy [5,6].

The present layout of a renewable hybrid system requires a suitable utilization of local resources, so the choice of the green energy mix is dependent not only on the local distribution of resources but also on whether they can be employed simultaneously in the same place [7]. In this context, it becomes imperative to explore creative and environmentally friendly options for improving energy efficiency, particularly through the hybridization of energy sources. However, the problem of how to hybridize multiple new energy systems within a single system is also a topical one, as the distribution and geographical distribution between the various energy sources are not strongly correlated.

1.2. Literature review

Hybrid Renewable Energy Systems (HRES) have shown great potential in multiple sectors, including transportation [8], industrial applications [9], and residential communities [10]. These systems address the twin objectives of energy security and environmental sustainability, while supporting the global transition from fossil fuels to renewable sources [11]. Given the intermittent nature of renewable energy, hybrid configurations combining multiple sources are particularly effective in meeting variable electricity demand throughout the day [12].

To implement HRES effectively, especially in microgrid contexts, thorough technical and economic analysis is necessary. Numerical modeling is commonly used to determine optimal system sizes and balance energy supply and demand. Various optimization approaches aim to minimize economic indicators such as fuel use, system degradation, environmental impact, Net Present Cost (NPC), and Levelized Cost of Energy (COE). HOMER Pro is one of the most widely adopted tools for these purposes, providing accurate simulations that account for daily and seasonal fluctuations in energy demand and resource availability. Additionally, HOMER's sensitivity analysis module evaluates the effects of varying parameters like fuel cost, interest rates, or renewable resource availability [13,14].

Several studies illustrate practical applications of HOMER Pro. Kapen et al. [15] assessed a hybrid energy system for Maroua, Cameroon, using solar photovoltaic panels, a biogas engine, a fuel cell, and an electrolyzer. They demonstrated that such systems could enable hydrogen production while improving energy access in the region, with average LCOH values of \$5.36/kg, \$3.145/kg, and \$0.30/kg depending on user consumption levels. El-Maaroufi et al. [16] developed a hybrid system to meet daily energy needs of 15,545.13 kWh in Zoumi, Morocco, combining solar (3877 kW), wind (1.65 MW), biogas (500 kW), batteries (5788 kWh), and converters (2224 kW). This configuration ensured a reliable supply even during fluctuations in renewable generation.

Further analysis by Yadav et al. [17] on three hybrid microgrid scenarios revealed that the second scenario, with a 1 % capacity shortage, struck a good balance between reliability and cost. In another context, Chamazkoti et al. [18] proposed a grid-connected hybrid system to power a wastewater treatment facility while producing green hydrogen. Their design, consisting of 1924 kW of solar panels, 1.5 MW wind turbines, 500 kW biogas generator, 1270 kW electrolyzer, and 578 kW converter, ensured 90 % reliance on clean energy. Similarly, Kumar et al. [19] optimized a system in rural eastern India using solar (600 kW), biomass (10 kW), and a 50 kW generator. The project demonstrated strong economic viability with a COE of \$0.222/kWh and a payback period of about 2 years. El-Sattar et al. [20] addressed electricity needs in remote areas with a peak demand of 420 kW using a combination of photovoltaics, biomass, fuel cells, and electrolyzers, successfully ensuring consistent supply and reduced emissions.

Building on these applications, other researchers have emphasized broader design tools and optimization techniques. The rising relevance of HRES is closely tied to global climate goals [21,22]. Tools such as HOMER Pro [23,24] and the System Advisor Model (SAM) [25] have been widely used, often coupled with optimization algorithms like

Particle Swarm Optimization (PSO) [26] and the Whale Optimization Algorithm (WOA) [27]. These methods enhance accuracy in system sizing while reducing losses and operational costs.

Economic performance indicators, especially the LCOE, are commonly used to assess system viability [23,28,29]. Some studies also report investment returns and payback periods, such as Nassar et al. [30] who identified a 7.54-year return in a Libyan case. Applications vary widely, from small-scale residential systems [31] to large institutional or national-scale projects [26], employing conventional sources like solar and wind [30,32], or advanced systems like fuel cells for water treatment [33] and pumped hydro storage [34].

To complement supply-side improvements, demand-side strategies are also crucial. Studies by Eteriki et al. [35] and Abdunnabi et al. [36] showed that residential efficiency enhancements could reduce annual energy use by over 50 %, reinforcing the need for integrated approaches to ensure sustainable and resilient energy systems.

1.3. Novelty, research questions, and contribution of the study

To address the persistent energy challenges faced by the IUT Douala campus, this study proposes an innovative hybrid energy system that integrates multiple sources photovoltaic (PV) solar panels, a biogas generator powered by human biomass, a diesel generator for backup, a battery storage system, and a grid connection. The system is designed to enhance energy autonomy, reduce dependency on fossil fuels, and optimize the use of locally available but underutilized resources such as human waste.

The techno-economic analysis, conducted using HOMER Pro software, compares two scenarios: (1) a conventional configuration reliant on diesel and grid electricity, characterized by high costs and carbon emissions; and (2) a hybrid system that synergistically combines solar energy and biogas production from human biomass, enabling a transition toward cleaner and more cost-effective energy.

This study provides a novel and context-specific contribution through the development and evaluation of a circular, sustainable energy model for an urban campus in Sub-Saharan Africa. The originality of the work lies in three main aspects: (i) the innovative use of on-campus human biomass as a primary feedstock for energy generation, an approach rarely explored in institutional microgrid design; (ii) the implementation of a circular economy model that simultaneously addresses energy security, sanitation, and waste management challenges; and (iii) the delivery of a replicable, data-driven case study that can inform energy and waste management strategies in similar environments.

To guide the investigation, the following research questions are posed:

1. What is the comparative techno-economic and environmental performance of the proposed PV/Biogas/Diesel/Grid hybrid system versus a conventional Diesel/Grid system for the IUT Douala campus?
2. To what extent can the integration of biogas from on-site human biomass reduce the levelized cost of energy (LCOE), operational costs, and greenhouse gas emissions?
3. How does implementing a circular economy model using human waste as an energy source affect the energy independence and long-term sustainability of an urban university in Sub-Saharan Africa?

By incorporating a neglected yet locally abundant resource into a resilient hybrid system, this study contributes to the discourse on sustainable energy transitions in resource-constrained settings and offers a model that may be adapted for other institutions facing similar energy and waste challenges.

1.4. Paper organization

The remainder of the article is structured as follows: Section 2 presents the datas and the methodology used, followed by Section 3, which discusses the modeling of hybrid renewable energy systems (HRES). Section 4 showcases the results along with related analyses, and Section 5 concludes the study, highlighting key findings and their implications.

2. Methodology

This study's methodology is based on a structured approach that incorporates load estimation, energy source evaluation, area selection, and techno-economic and environmental analysis prior to optimizing hybrid systems based on two hybrid scenarios: (1) Diesel Generator/Grid and (2) PV/Diesel Generator/Biogas Generator/Battery/Grid. HOMER Pro software was used to produce technical, financial, and environmental outcomes, as the flow chart in Fig. 1 below illustrates.

With the use of HOMER Pro software, users may simulate various energy source, storage system, and load combinations to evaluate various energy configurations and identify the most dependable and economical installations. The program is intended to replicate realistic situations by taking into account variations in wind speed, sun radiation, and other pertinent factors. It meets certain dependability requirements while optimizing the design according to an objective function that reduces expenses or emissions. This method works well for examining scenarios including the integration of renewable energy sources, independent microgrids and connected to the grid systems in isolated areas. Additionally, it covers a variety of metrics, such as the techno-economic parameters pertinent to the study, environmental variables influencing the availability of clean energy supplies, and the performance of system components.

2.1. Area selection

In order to address the energy requirements of the University

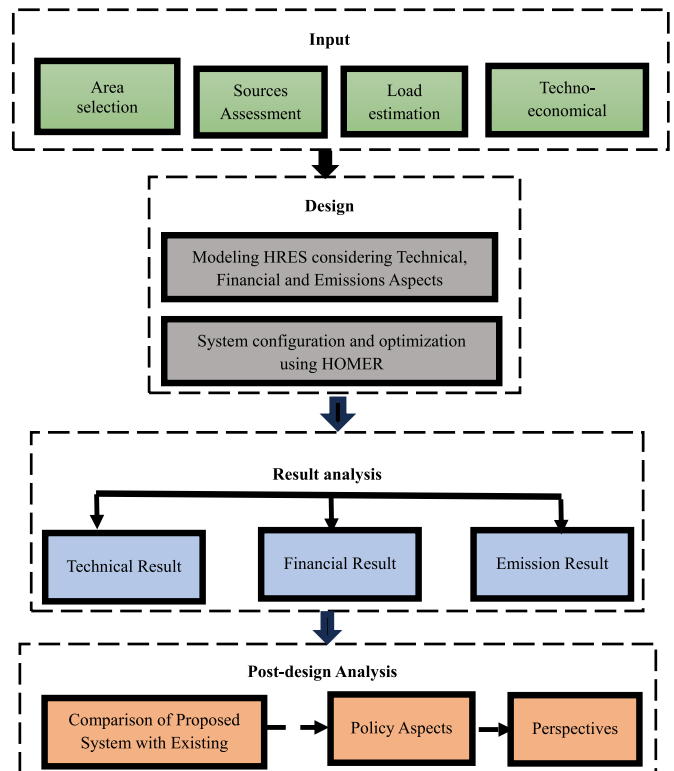


Fig. 1. Workflow with HOMER software simulation process.

Institute of Technology (UIT) administrative block at the University of Douala, this research will conduct a technical and financial analysis of an appropriate hybrid system. The campus is situated on the banks of the Wouri River in Douala, the commercial center of Cameroon. Its coordinates are 4°3'53.77" N and 9°41'15.41" E. Its tropical climate means that the city has four distinct seasons: an abundant rainfall from May to October, a brief dry time from November to December, a short-wet time in January and February, and a long dry season from March to May. The temperature is about 27.5 °C, and there is an average of 963.7 mm of rainfall each year. The campus, which is situated in a semi-urban area in Douala 5e (Fig. 2), has brittle energy networks that are frequently interrupted by power outages, and the administrative block, which is the school's pilot center, is having problems because of poor energy management and a significant reliance on diesel generators, even though the area has renewable resources.

2.2. Electrical load estimation

The Douala UIT administrative block's electrical demand is estimated by carefully examining the building's equipment and how much electricity it uses. To achieve this, we calculated the overall load of a number of frequently used electrical items, including air conditioners, computers, televisions, and refrigerators, by looking at their hourly usage.

We determined the highest hourly requirement as a function of the total amount of devices and their usage time based on the devices that were detected and their corresponding power ratings (in watts). A 1700 W air conditioner, for instance, that is operated constantly during business hours would significantly affect demand. Similarly, even though they use less electricity, devices like printers, computers, and microwaves nevertheless make a substantial contribution to overall consumption. Table 1 displays the comprehensive estimation of the electrical consumption for every hour of the entire day that we were able to acquire using these data. Fig. 3 displays the profile of the daily load. The precision of this hourly assessment is crucial, given that the time-dependent nature of the load data significantly influences the optimization process within HOMER Pro. It dictates the required capacity of the photovoltaic (PV) array to handle daytime peak demands, guides the

sizing of the battery storage system to ensure grid stability and provide buffering, and informs the generator dispatch strategy to maintain a consistent and reliable power supply amidst variable demand.

2.3. Sources assessment

2.3.1. Solar

Significant solar potential is found in the area, especially during the dry season from November to March, according to the study's meteorological data, which was taken from NASA's Surface Meteorology and Solar Energy Database. However, from May to October, when it rains, solar potential drops. Fig. 4 displays an average of hourly profiles of site transparency index and solar irradiance that were gathered in 2022. The average yearly solar radiation in Douala is 4.2 kWh/m²/day. Irradiation peaks in January at 5.05 kWh/m²/day and troughs in August at 3.85 kWh/m²/day. The transparency index's monthly averages range from 0.32 in August (extremely overcast) to 0.55 in January (very sunny), suggesting that this research has a comparatively considerable opportunity for solar energy utilization.

2.3.2. Temperature

A solar system's efficiency and ability to produce electricity are significantly impacted by temperature. Warmer climates can, in fact, cause solar panels to become less productive and, as a result, less able to produce electricity. The research area's average yearly temperature was found to be 25.93 °C. Fig. 5 displays daily temperature changes each month.

2.3.3. Biogas

Human fecal matter represents a significant source of biomass that can be converted into energy. Through processes such as anaerobic digestion, it is possible to produce biogas, which can then be used to generate electricity and heat. This method of energy recovery becomes even more efficient when integrated with heat recovery systems, offering a sustainable alternative to fossil fuels, especially when combined with other renewable energy sources [37].

At the University Institute of Technology (UIT) in Douala, the availability of human fecal biomass is intrinsically tied to campus occupancy. To realistically capture seasonal variations, a dynamic monthly biomass availability profile was established, assuming an average generation rate of 120 g of fecal matter per person per day.

This model categorizes the year into four distinct levels of biomass supply, aligned with the academic calendar and holiday breaks:

- A peak academic load of 180 kg/day-corresponding to approximately 1,500 individuals-is assumed for January, February, March, May, June, October, and November.
- A minimum availability of 36 kg/day-reflecting a significantly reduced presence of about 300 individuals-is applied to the core holiday months of July and August.
- Transitional periods are represented by two intermediate levels: 146.4 kg/day in April and December, which include short Easter and Christmas breaks, and 108 kg/day in September, accounting for the gradual return of students after summer vacation.

This variable biomass input, illustrated in Fig. 6, was integrated into the HOMER Pro modeling framework to simulate the system's performance under realistic annual fluctuations. Associated operational costs for waste collection and management were also incorporated into the economic analysis.

In this study, this volume of waste was analyzed using HOMER Pro software to optimize a hybrid energy system [38,39]. The operational costs related to the management and collection of this waste, including potential leakage losses, were factored into the system's cost analysis. Biogas yields were estimated within a range of 0.70 m³, with an associated energy production cost of approximately 1 kWh [3].

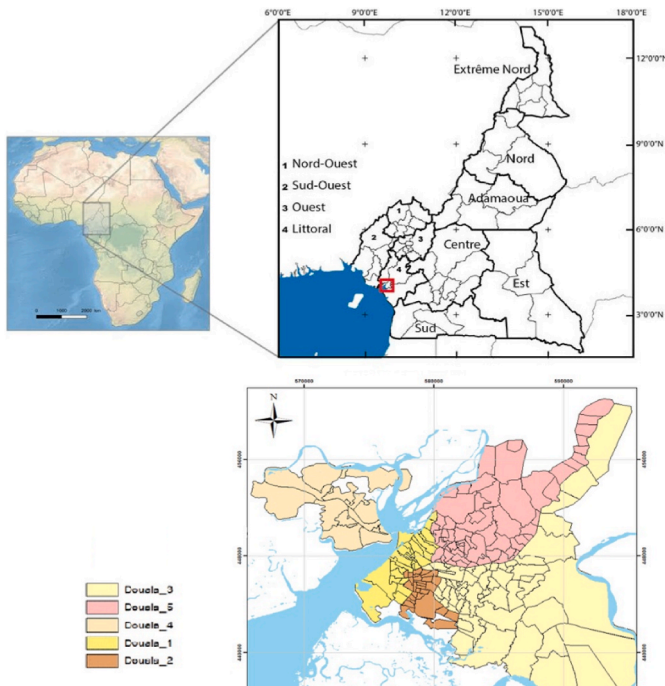
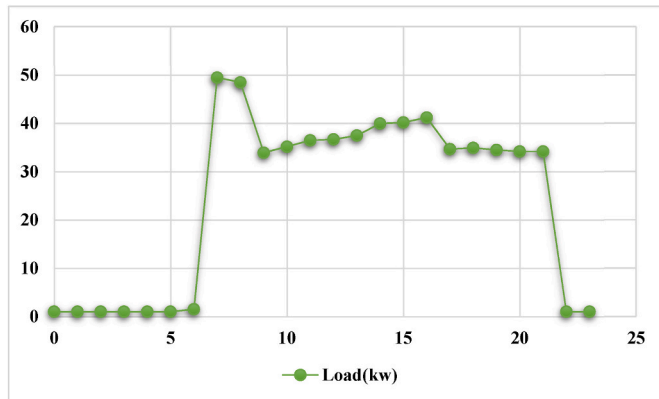


Fig. 2. Geographical location of the study area.

Table 1

Detailed daily electrical load estimation for the administrative block.

Time	Led (11 W)	Air conditioner (1700 W)	Microwave oven(1000 W)	TV (100 W)	Coffee maker (750 W)	Lap Top (75 W)	Fridge (250 W)	Desktop (350 W)	Phone Charger (5 W)	Printer (250 W)	Camera (20 W)	Energy Consumed (W/h)
01:00	12			1			3				1	1,002
02:00	12			1			3				1	1,002
03:00	12			1			3				1	1,002
04:00	12			1			3				1	1,002
05:00	12			1			3				1	1,002
06:00	12			1			3				1	1,002
07:00	12			1			3			2	1	1,502
08:00	28	15	3	3	15	12	3	18	10	5	1	49,448
09:00	28	15	3	3	15	12	3	18	15	1	1	48,473
10:00	28	15		3		12	3	18		0	1	33,898
11:00	28	15		3		12	3	18		5	1	35,148
12:00	28	15		3		12	3	18	12	10	1	36,458
13:00	28	15	2	3		12	3	18		3	1	36,648
14:00	28	15	2	3		12	3	18	10	6	1	37,448
15:00	28	15		3	8	12	3	18	7		1	39,933
16:00	28	15		3	8	12	3	18		1	1	40,148
17:00	28	15		3	8	12	3	18	8	5	1	41,188
18:00	28	15		3		12	3	18		3	1	34,648
19:00	28	15		3		12	3	18		4	1	34,898
20:00	28	15		3		12	3	18	12	2	1	34,458
21:00	28	15		3		12	3	18		1	1	34,148
22:00	28	15		3		12	3	18		1	1	34,148
23:00	12			1			3				1	1,002
00:00	12			1			3				1	1,002
Total												580,608

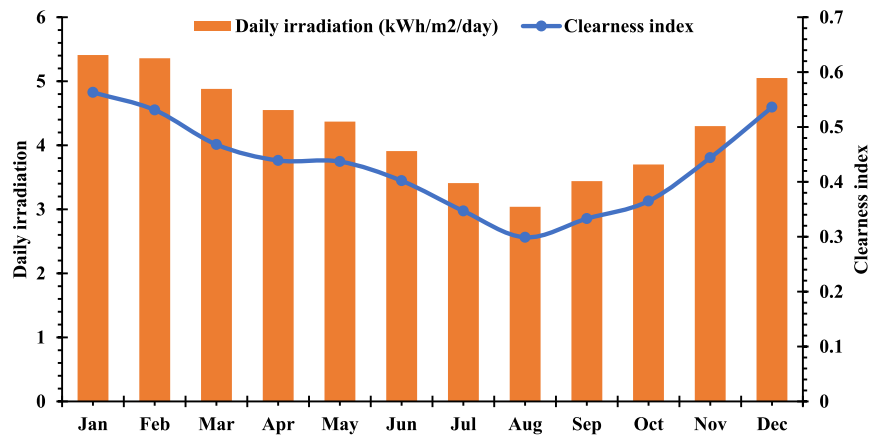
**Fig. 3.** Hourly energy consumption trends in the study area.

3. Hybrid system modeling and components

The optimization was performed using HOMER Pro software. The primary objective function of the simulation is to minimize the total Net Present Cost (NPC) of the system, which consequently minimizes the Levelized Cost of Energy (LCOE). This optimization is subject to the following key constraints:

- The system must meet the defined hourly electrical load throughout the year without any load rejection (0 % capacity shortage).
- The operation of each component (PV, generators, battery) must respect its technical limits, such as minimum load ratios for generators and depth-of-discharge limits for batteries.
- Energy generation from renewable sources is constrained by the real-world availability of solar radiation and biomass feedstock.

HOMER Pro conducted the analysis of two different hybrid systems, namely, PV/Biogas/Diesel/Storage/Grid (Fig. 7a) and the actual

**Fig. 4.** Hourly solar irradiance profile for the study area.

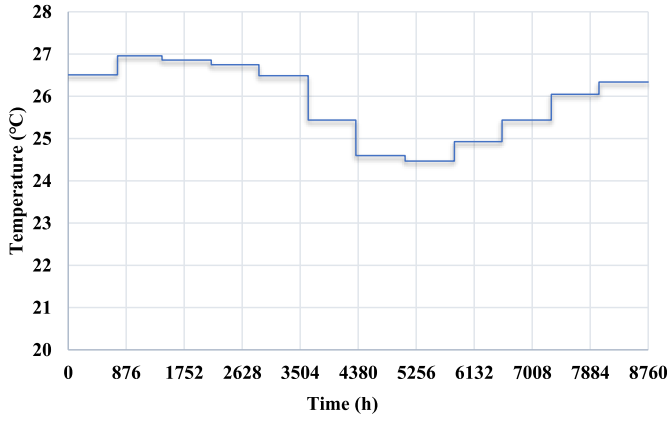


Fig. 5. Ambient temperature data for the study area.

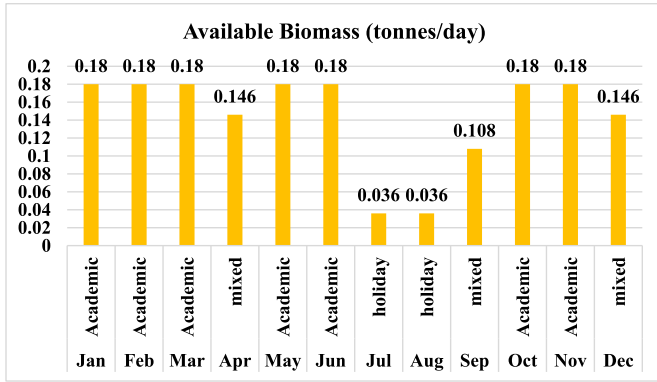


Fig. 6. Monthly biomass availability profile accounting for seasonal campus population changes.

configuration Diesel/Grid (Fig. 7b), in this manuscript. The properties of several components are also outlined in this section.

3.1. Solar

The PV solar panels create electricity based on the solar irradiation measured by HOMER Pro. This state is important in the modeling of hybrid systems since it takes into account a variety of elements such as weather and time of day. Equation (1) [40] may be used to calculate the energy output of a PV solar panel.

$$P_{PV} = y_{PV} \times f_{PV} \times \left(\frac{\bar{G}_T}{\bar{G}_{T,STC}} \right) \times [1 + \alpha_p (T_C - T_{C,STC})] \quad (1)$$

T_C is the cell temperature, which is determined by Expression (2).

$$T_C = T_{amb} + (0.0256 \times G) \quad (2)$$

In this case, y_{PV} stands for the PV generator's rated output under standard test conditions [kW], f_{PV} for the PV derating factor [%], $\bar{G}_T$ for the solar radiation incident on the PV generator at the current time step [kW/m²], $\bar{G}_{T,STC}$ for the radiation under standard test conditions [1 kW/m²], α_p for the temperature coefficient of power [%/°C], T_C for the PV cell temperature at the current time step [°C], and $T_{C,STC}$ for the PV cell temperature under standard test conditions [°C].

For the purpose of this simulation, a standard monocrystalline silicon photovoltaic (PV) module was selected, given its favorable trade-off between efficiency and cost [41]. This technology is particularly well-suited to the regional context and the application considered in this study. The HOMER program was used to develop the system. The

utilized panel has a 25-year lifespan. The cost of a 1 kW panel is \$600, the cost of renewal is \$600, and the cost of operation and maintenance is \$10 per year. The derating factor for the panel to be utilized in the system is set at 80 % to ensure that production losses due to dirt, temperature, shadow, and age do not surpass 80 %. Table 2 displays the solar panel's technical parameters.

3.2. Biogas modeling

HOMER Pro simulates the conversion of human fecal biomass to electricity through a literature-based, multi-step pathway. The daily inflow of wet waste is first reduced to its energy-bearing fraction: the model assumes 25 % total solids (TS) within the wet mass, of which 85 % are volatile solids (VS) that can undergo anaerobic digestion. Applying a conservative yield of 0.28 m³ of biogas per kilogram of VS produces an estimated gas volume whose methane (CH₄) share is set at 60 % [43–46].

This methane stream fuels a 500-kW internal-combustion engine (ICE). Electrical output is calculated from the methane's lower heating value (≈ 9.94 kWh m⁻³) and an engine efficiency of 30 %, parameters typical for medium-scale biogas gensets. To safeguard the engine and comply with emissions norms, the raw biogas is pre-treated: moisture is removed (dehumidification) and hydrogen sulfide is stripped out via iron-oxide filtration. Both capital and operating expenses for this conditioning stage are embedded in HOMER Pro's cost inputs [47].

By combining these technical and economic factors, the model can track how seasonal fluctuations in biomass supply translate into reliable, time-resolved electricity generation profiles [48].

3.3. Diesel

The power infrastructure for the study includes a diesel generator. A fuel-powered generator was installed, adhering to the HOMER program's requirements for an Auto size Genset generator. The cost of replacing a diesel generator was similarly set at \$500 per kilowatt-hour. At the same time, an estimate of \$0.030 per hour was made for operating and maintenance expenses. The generator has a lifetime of 15,000 h. The generator requires 5.36 L of diesel fuel per hour to run. Diesel for the diesel generator costs \$1.18 per litre when using the Bank of Central African States' (BEAC) January 20, 2025, currency rate (\$1 = 608,59FCFA). The following Equation (3) [40] may be used to determine how much fuel a diesel generator uses:

$$F_{dg} = B_g \times P_{dg} + A_g \times P_{dg-out} \quad (3)$$

Table 3 displays the emissions and fuel characteristics. P_{dg-out} represents the DG output power, P_{dg} indicates the DG rated power, and A_g and B_g are the fuel's consumption constant coefficient constants, which are roughly 0.246179 l/kW and 0.08415 l/kW, respectively.

3.4. Storage

In this system configuration, the battery pack is integrated with a targeted operational role rather than for bulk energy storage. Its principal function lies in power stabilization and short-term bridging rather than in covering prolonged energy deficits. The battery is sized to deliver instantaneous power, effectively serving as a spinning reserve to enhance system stability in two critical situations: (1) bridging the brief interval between a grid outage and the startup/synchronization of the diesel generator, thereby avoiding load interruptions; and (2) mitigating fluctuations in the photovoltaic (PV) output, thereby enhancing the quality and consistency of the power delivered to the load. A charge controller is implemented to safeguard the battery from overcharging. This strategy allows for a cost-effective storage solution, optimizing capital investment while maintaining high system reliability through its complementary function alongside the diesel generator. The battery

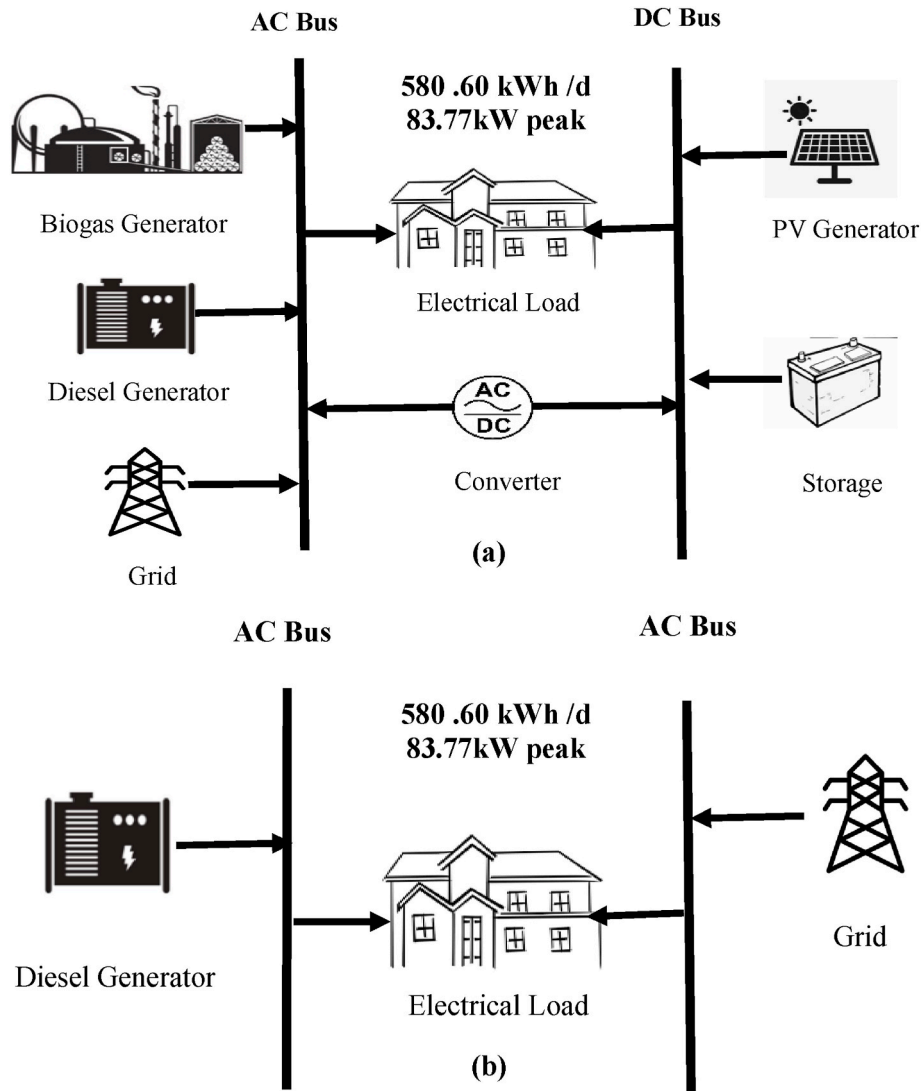


Fig. 7. Hybrid energy system (a) and the conventional system (b).

Table 2
Technical specifications of solar modules [42].

Component	ITEM	Value
Solar panel	Panel power rating (kW)	1
	Normal operating cell temperature ($^{\circ}\text{C}$)	25
	Temperature coefficient of power ($\%/^{\circ}\text{C}$)	-0.45
	Derating factor (%)	80
	Efficiency at STC (%)	18.5
	Degradation factor (%)	80
	Ground reflectivity (%)	20
	Duration (years)	25

capacity can be calculated using Equation (4).

$$B_C = V \times Q \quad (4)$$

where B_C is the capacity of the battery, V is its voltage, and Q is its electrical charge.

The total energy needed for powering the storage device (kWh), E_i , may be written as follows in Equation (5) [49]:

$$E_i = \frac{C_i}{\eta_{\text{charger}}^i} \times (1 - \text{SOC}_i) \quad (5)$$

Table 3
Emissions and characteristics of diesel fuel.

Component	ITEM	Value
Diesel generator	Lower calorific value (milligrams/kilogram)	43.2
	Concentration (kilogram/meter cube)	820
	Carbon level (%)	88
	Sulphur level (%)	0.4
	Carbonic oxide (g/L fuel)	16.5
	Nitrogen oxides (grams/litre combustible)	15.5
	Uncombusted hydrocarbons (grams/litre combustible)	0.72
	Particle size (grams/litre combustible)	0.1
	Sulphur dioxide (%)	2.2

where η_{charger}^i is the battery charger's efficiency, C_i is the battery's capacity, and SOC_i is the vehicle battery's state of charge (SOC) expressed as a percentage (%).

Equation (6) [50] may be used to determine the battery's immediate state of charge:

$$\text{SOC}(t) = \text{SOC}(t-1) \times \left(1 - \frac{\sigma \times \Delta t}{24}\right) + \left(\frac{I_{\text{bat}(t)} \times \eta_{\text{bat}} \times \Delta t}{C_{\text{bat}}}\right) \quad (6)$$

where $I_{\text{bat}(t)}$ is the battery current at time t , η_{bat} is the battery charge

efficiency, C_{bat} is the battery bank's capacity, and SOC ($t-1$) is the battery's self-discharge rate at ($t-1$) hours. Table 5 displays the technical and economic parametric values of the aforementioned technical components.

3.5. Converter

A bidirectional converter is mostly used in the research work. This sort of converter may function as both an inverter and a rectifier, depending on the system's requirements.

The bidirectional converter selected has an efficiency of 95 % for both inversion and rectification modes [41].

The essential elements of the hybrid energy system are enumerated in Table 4 together with their lifetime, operation and maintenance (O&M) expenses, replacement costs, and capital investment. Photovoltaic panels, a diesel generator, a biogas generator, batteries, and a converter are among the parts. Because of mechanical degradation, generators have lower operating lifespans than photovoltaic panels. The lifespans of batteries and converters are moderate, and they require particular care. This cost structure influences the hybrid energy system's overall performance and financial feasibility by highlighting the harmony between initial investments, maintenance needs, and durability.

3.6. Grid

The State power Board, which is managed by the Cameroonian government's Department of Power, is in charge of providing power to the public at a cost of around \$0.1 per kilowatt-hour (kWh). However, the electricity supply is afflicted by frequent outages, making it noticeably unstable. The distribution station is near to residential areas, however a lack of detailed information about power system disruptions is a major issue. The proposed HRES uses a simulation-centric technique to assess dependability. This simulation is based on the frequency of power outages, which occur 365 times a year on average. Each outage requires roughly 2 h of repair time, as seen in Fig. 8. During these outages, all key parties and the grid are unable to exchange electrical energy.

3.7. Economic modeling

3.7.1. Net Present Cost (NPC)

The NPC of an element is determined as the difference among its current worth of expenses over the course of the project and its current value of returns over the same period. Equation (7) can be used for the assessment procedure, as detailed in sources [35,36].

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i, R_{proj})} \quad (7)$$

The annualized cost comprises yearly operation and maintenance (O&M), emissions, and fuel (generator) costs. $CRF(i, R_{proj})$ calculates a return on capital (CRF). The yearly effective discounted rate [%] and

Table 4

Techno-economic characteristics of the hybrid system components.

Components	Capital investment	Replacement Cost	O&M Cost	Life time	Refs
Photovoltaic panel	600 \$/kW	600 \$/kW	10 \$/kWh	25 years	[15]
Diesel generator	500 \$/kW	500 \$/kW	0.03 \$/kWh	15000 h	[51]
Biogas generator	500 \$/kW	450 \$/kW	0.03 \$/kWh	20000 h	[3]
Battery	300 \$/kW	300 \$/kW	10 \$/kWh	10 years	[51]
converter	300 \$/kW	300 \$/kW	1 \$/kWh	15 years	[7]

Table 5

Optimization Results of the Proposed Hybrid System and the conventional system.

Item	Component	unit	Hybrid system	Conventional system
System configuration	Solar panels	kW	107	–
	Batteries	Quantity	8	–
	Biogas generator	kW	500	–
	Diesel generator	kW	93	93
Cost	Dispatch strategy	LF or CC	LF	LF
	LCOE	\$/kWh	0.07563	0.1597
	NPC	\$	300,661	437,605.20
Energy generation	Solar panels	kWh/yr	134,325	–
	Biogas generator	kWh/yr	84,793	–
	Diesel generator	kWh/yr	13,242	24,281
	generator			

project duration R_{proj} are given by the equation below. The Capital Recovering Factor (CRF) is calculated using Equation (8) [52,53,54].

$$CRF(i, R_{proj}) = \frac{i(1+i)^{\tau}}{(1+i)^{\tau} - 1} \quad (8)$$

The research considers the plant's life span to be 25 years, represented by the symbol τ .

The net discounted rate (i) in the HOMER Pro software has been defined as the real interest rate used to convert expenses from annualized to in-time values. HOMER calculates the net rate of interest using Equation (9) [52].

$$i = \frac{i - f}{1 + f} \quad (9)$$

3.7.2. Cost of energy (COE)

One of the key metrics in the energy industry is the Levelized Cost of Energy (LCOE), sometimes simply called the Cost of Energy (COE). In essence, it tells you the average price you pay for each kilowatt-hour of useable electricity that a system delivers over its entire lifespan.

In our analysis, we rely on HOMER Pro to compute the LCOE using a widely accepted industry method. The software divides the system's total annualized cost ($C_{ann,tot}$) by its total yearly energy output (E_{gen}). To get the annualized cost, HOMER Pro multiplies the project's Net Present Cost (NPC) by a Capital Recovery Factor (CRF), as outlined in equation (10) [55,56].

$$COE = C_{NPC} \times CRF \left/ \sum_{t=1}^{8760} E_{gen} \right. (\$/kWh) \quad (10)$$

where E_{gen} is the sum of the energy produced by the microgrid (MG) and the grid. Equation (11) is used to obtain it.

$$E_{gen} = E_{gen_{MG}} + E_{gen_{grid}} \quad (11)$$

Whereas $E_{gen_{grid}}$ represents the energy produced only by the grid, $E_{gen_{MG}}$ represents the energy produced by the microgrid.

3.7.3. Emission modeling

Environmental or greenhouse gas emissions associated with non-renewable energy source utilization play a major role in air pollution and climate change. Shifting to renewable or clean energy sources (solar, wind, biogas, etc.) helps mitigate these emissions. Hybrid systems significantly reduce these harmful outputs while enabling cleaner and more sustainable power generation by decreasing the reliance on fossil fuels. In particular, incorporating hydrogen production processes into these systems helps reduce greenhouse gas emissions. In this study,

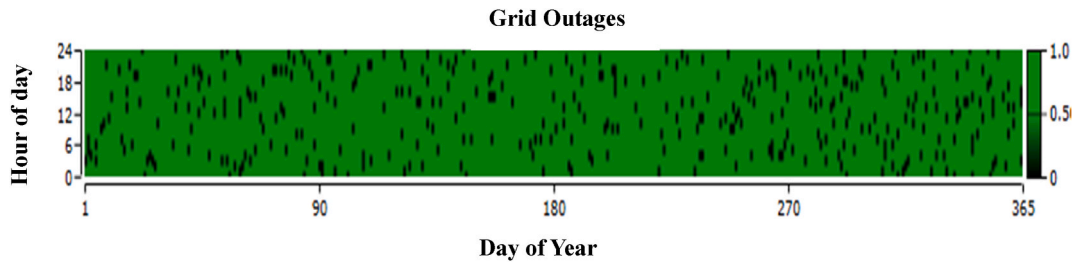


Fig. 8. Annual grid outage profile.

although the emissions of system generated components are not considered, only operational emissions depending on the operating conditions and optimization of the system are analyzed. The main source of operational emissions is the biogas generator, while the intermediate source is the operation of the grid system Equations 12 and 13 determine the system's CO₂ emission value, which is as follows [37].

$$CO_{2emission} = HRES(kWh) \times 0.318 \left(\frac{kg_{CO_2}}{kWh} \right) \quad (12)$$

$$CO_{2reduction} = BG_P(kWh) \times 0.43 \left(\frac{kg_{CO_2}}{kWh} \right) \quad (13)$$

In addition to analyzing emission volumes, this study also quantifies the financial cost of environmental damage associated with carbon dioxide (CO₂) emissions from electricity purchased from the grid. This cost (C_{CO₂}) is calculated using the following formula (14) [56,57]:

$$C_{CO_2} = F_{CO_2} \times E_{grid} \times \phi_{CO_2} \quad (14)$$

Where E_{grid} is the total energy purchased from the grid annually (kWh/year), F_{CO₂} is the CO₂ emission factor of the power generation system, considered here as 0.318 kg CO₂/kWh, and ϕ_{CO_2} is the social cost of carbon, estimated at \$70/ton of CO₂ [58].

This analysis allows for the translation of environmental impact into an economic indicator, thereby strengthening the comparative assessment of the two systems.

3.8. Assumptions, limitations, and uncertainties

This study relies on several assumptions and is subject to specific limitations and uncertainties that may affect the interpretation of results. The electrical load was estimated from typical usage patterns within the administrative block rather than continuous monitoring. Solar resource data were drawn from the NASA database, which may differ slightly from local measurements [59]. Biomass availability was calculated using standard estimates of daily fecal matter production and typical biogas yields, though actual values may vary. Cost parameters for system components were derived from literature and market estimates, which are inherently subject to fluctuation. Furthermore, as a techno-economic simulation, the study does not account for potential social resistance to human waste reuse or assess stakeholder perceptions. It also excludes life-cycle environmental impacts such as emissions from component production and disposal. Finally, key financial indicators are sensitive to fuel prices, electricity tariffs, and interest rates, introducing uncertainties despite the sensitivity analyses performed in HOMER Pro.

4. Results and discussions

4.1. Optimal system configuration

This section compares hybrid and conventional energy systems, with an emphasis on system layout, cost metrics, energy generating efficiency and emission analysis. The hybrid system combines 107 kW of solar

panels, 8 battery units, and a 93-kW diesel generator, whereas the traditional system just uses a diesel generator of the same capacity. The hybrid system is more cost-effective, with a Levelized Cost of Energy (LCOE) of \$0.07563/kWh, which is much less than the traditional system's \$0.1597/kWh. Additionally, its Net Present Cost (NPC) is decreased to \$300,661, as opposed to \$437,605.20 for the typical system. In terms of energy generation, the hybrid system generates 134,325 kWh/year from solar and 84,793 kWh/year from biogas, significantly reducing diesel use to 13,242 kWh/year. Table 5 highlight the hybrid system's superior economic and energy efficiency, demonstrating the benefits of renewable integration.

4.2. Energy performance analysis

4.2.1. Monthly energy balance

The monthly power generation of the suggested hybrid system (Fig. 9 b) and the existing system (Fig. 9 a) is contrasted in the figure. In order to compensate for grid outages, the current system uses a diesel generator in addition to the public grid. The hybrid system, on the other hand, combines storage, a diesel generator, biogas generation that uses human biomass as an energy source, and photovoltaic (PV) panels, all of which are connected to the public grid. Figure analysis shows that the existing system and the suggested hybrid system have different energy source distributions. Monthly energy output in Fig. 9 a is mostly dependent on the public grid, which provides around 88.9 % of the overall supply, with the diesel generator contributing 11.1 %. With sporadic reliance on fuel to make up for service outages, this distribution shows a heavy reliance on the grid.

On the other hand, Fig. 9 b emphasizes energy diversity with the incorporation of the biogas generator and solar panels. Approximately 30.8 % of this design comes from the public grid, whilst 40 % comes from solar energy and 25.3 % comes from biogas. Diesel generator usage is still quite low, at less than 3.94 %, which suggests a considerable decrease in the use of fossil fuels. The effectiveness of the hybrid system, which uses renewable resources to provide a steadier supply that is less reliant on variations in the public grid, is shown by this redistribution of energy sources.

4.2.2. Photovoltaic system performance

Fig. 10 depicts a time-based graph of the solar system's energy production vs time of day and day of year. This graph illustrates the daily and seasonal fluctuation of solar output. The distribution of energy generated shows that production is peaked between 6 a.m. and 6 p.m., which corresponds to sun exposure throughout the day. Production peaks at midday, when sunlight is at its most intense, and there is no production at night. The variation seen throughout different days of the year indicates the effect of weather and seasonal variations on solar energy collection.

With an annual total production of 134,325 kWh and a nominal capacity of 107 kW, the PV system contributes significantly to the hybrid system's energy supply. With a particular yield of 1,256 kWh/kW, this generator is comparatively efficient in terms of energy performance; the system's \$64,161 initial investment cost and \$1,069 annual

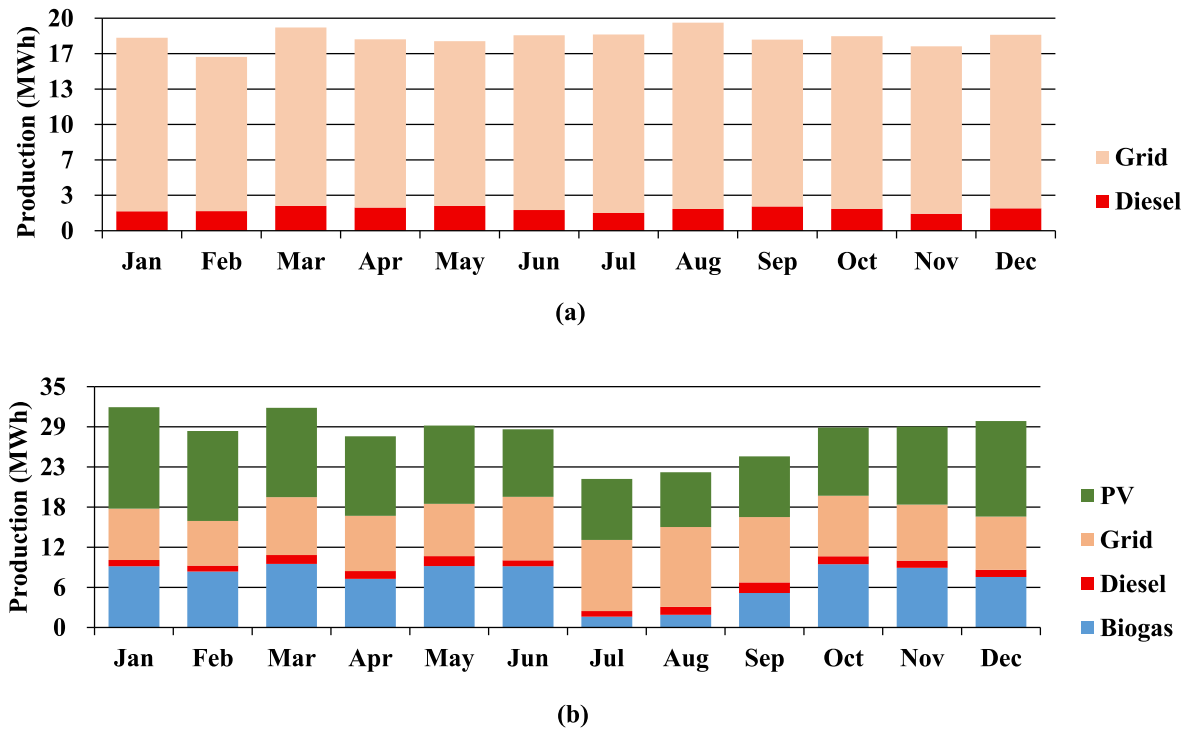


Fig. 9. Monthly average electricity production of both systems.

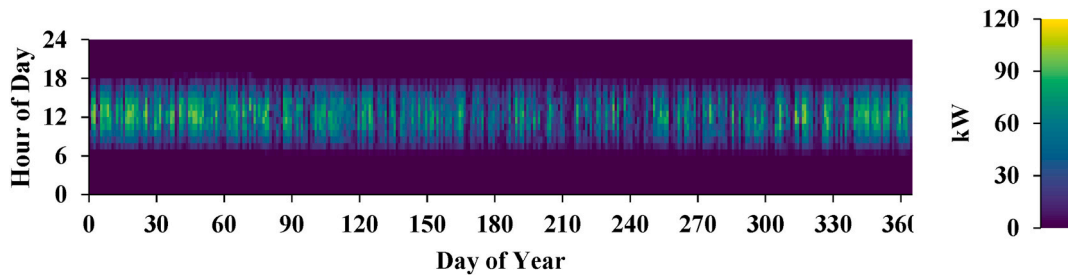


Fig. 10. Photovoltaic (PV) power output in the hybrid system.

maintenance cost make it a cost-effective long-term solution; its Levelized Cost of Energy (LCOE) is \$0.0449/kWh, making it an affordable energy source when compared to other production methods. More than half of the energy requirements are met by photovoltaics, which have a penetration rate of 63.4 % in terms of integration inside the hybrid system. To guarantee a steady power supply, particularly during times of low sunshine, this source must be combined with biogas, a diesel generator, and energy storage because to the daily and seasonal variations seen in the graph.

4.2.3. Diesel generator performance

In each of the systems under study, the diesel generator, which has a nominal capacity of 93 kW, acts as a backup. However, there are substantial differences in how it is used between the hybrid and conventional systems, which affects environmental effect, operational costs, and fuel consumption (Fig. 11). The generator in the traditional system runs for around 739 h annually, which indicates a high load and a significant dependence on diesel to make up for grid irregularities. As a consequence, more than 8,353 L of gasoline are used each year, which raises operational expenses and leaves a large carbon imprint. The consumption of diesel generators is greatly decreased by combining solar photovoltaics with biogas. Compared to the traditional system, it now only runs 455 h a year, which is 38 % fewer. Every year, fuel usage

drops by around 3,613 gallons, which lowers CO₂ emissions and operating expenses. Long-term profitability is increased as a result of the generator's lifespan being increased from 20.3 to 33 years due to the reduction in mechanical workload.

These findings show that incorporating renewable energy into a hybrid system significantly lowers the diesel generator's usage. Reducing its use also lowers maintenance expenses and emissions of pollutants in addition to fuel consumption. Diesel turns into a fallback option, mostly utilized during periods of high demand or when output from solar and biogas is minimal. While reducing its negative effects on the environment and the economy, this move toward more moderate diesel use guarantees increased system sustainability.

The diesel generator's limited annual runtime (455 h) is explained by its designated role within the "Load Following" dispatch strategy of the grid-connected hybrid system. Priority is given to the PV array and biogas generator, followed by grid electricity, which is more cost-effective than diesel. As a result, the diesel generator operates only during grid outages that coincide with insufficient renewable generation. This hierarchical supply structure-renewables first, grid second, diesel last-ensures high system reliability while minimizing fuel use, operational costs, and wear, reinforcing the generator's role as an emergency backup rather than a primary source.

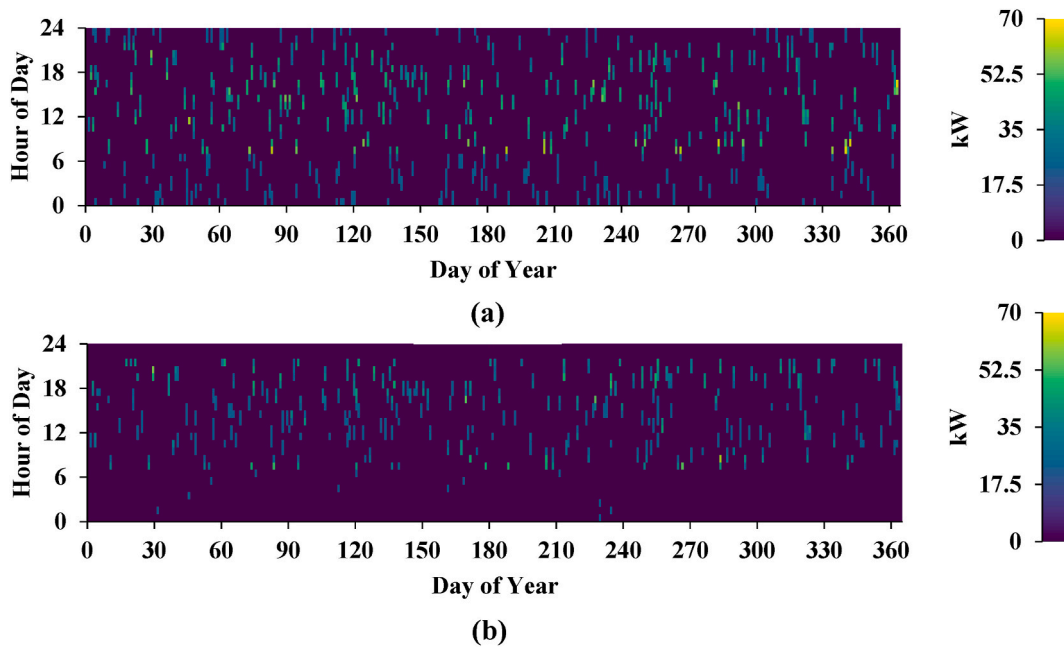


Fig. 11. Power output from the diesel generator.

4.2.4. Biogas generator performance

The hour and seasonal variation in energy output from the biogas the generator, which runs on biogas as fuel, is depicted in Fig. 12. This graph illustrates the biogas generator's consistent operation, with the generation of power being distributed year-round. In contrast to photovoltaics, whose output varies according to sunshine, the biogas generator produces power consistently regardless of the weather. The output is seen to begin at regular times every day, indicating a well-planned and efficient operation intended to satisfy the hybrid system's energy requirements.

The biogas generator demonstrates strong performance from both economic and energy perspectives. With a rated capacity of 750 kW, it consistently delivers approximately 84,793 kWh annually, highlighting the effectiveness of the biomass recovery process through the utilization of 52.6 tons of biogas per year. A key advantage of biogas is its independence from fossil fuel procurement, leading to significantly lower operational costs compared to the diesel generator, which has a marginal cost of \$0.052/kWh. Additionally, with an estimated lifespan of 65.1 years, the biogas unit offers a highly sustainable and cost-effective long-term solution.

The seasonal variation in biomass availability plays a notable role in the system's performance. As depicted in Fig. 6, biogas production drops during the 2.5-month academic break due to reduced campus occupancy. HOMER simulations show that the system compensates for this decline by increasing reliance on grid electricity, while the diesel generator remains in standby mode for grid interruptions. This adaptive

strategy maintains year-round power continuity and prioritizes renewable sources during active academic periods. Although seasonal variability slightly increases annual grid dependence compared to a constant biomass scenario, the hybrid system continues to offer superior techno-economic and environmental advantages over conventional configurations.

4.2.5. Battery and grid interaction

The battery's level of charge is shown in Fig. 13 as a function of day and time of day. With sporadic discharge times depicted as blue-green patches, the graph indicates that the battery is primarily charged at a high level (around 100 %) throughout the year. This suggests that although the battery is being used well, it frequently runs at maximum capacity, which restricts its ability to regulate energy.

The battery storage system plays a crucial role in power stabilization rather than extended energy shifting, a design choice reflected in its final specifications (Fig. 13). With an autonomy of just 13 min and a capacity of 8.01 kWh, the battery primarily serves as a short-term backup to ensure uninterrupted supply during grid outages or delays in renewable availability, bridging the gap until the 93-kW diesel generator activates. Its consistently high state of charge throughout the year confirms that it is used for brief, intermittent events rather than deep, routine cycling. Although its lead-acid chemistry results in notable energy losses (22.3 % annually), the system strikes a cost-effective balance—leveraging a compact, low-cost battery for immediate stabilization, while deferring long-duration reliability to the diesel generator. This synergy maintains

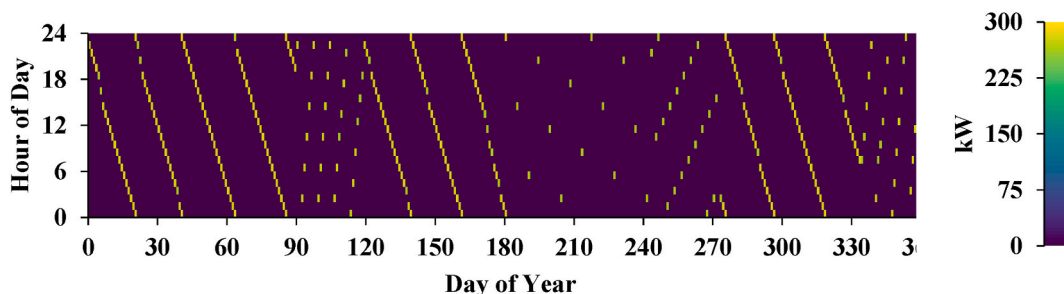


Fig. 12. Power output from the biogas generator.

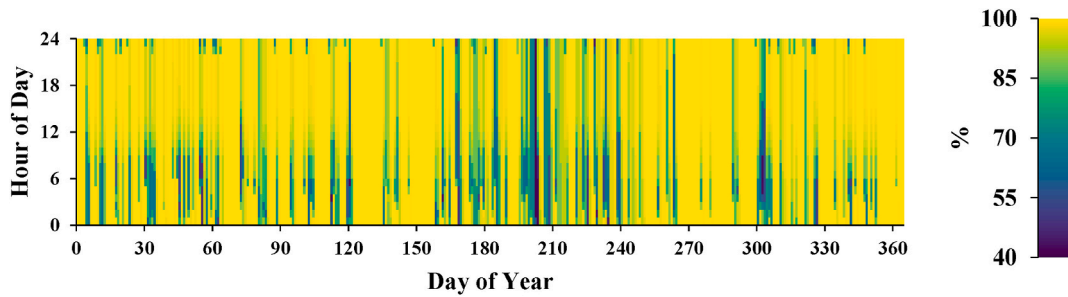


Fig. 13. Annual battery state of charge profile.

operational resilience during periods of low renewable output.

Data analysis shows that the hybrid system and the traditional system differ significantly in how energy is bought and sold to the grid. The hybrid system lessens reliance on the grid by consuming a significant amount of energy from renewable sources (biogas and photovoltaics). The fact that excess energy is sold back to the grid, totaling 95,582 kWh year or around 92 % of the energy purchased, is one of the main distinctions from the traditional approach. By balancing consumption, this energy reinjection process lowers the cost of purchasing power.

The grid provides all of the energy used in the traditional system; renewable sources do not provide any compensation. With an energy bill of \$19,374-more than three times the cost of the hybrid system-the yearly consumption is almost twice as high as with the hybrid system. The system is more susceptible to changes in power prices and the possibility of outages because of its complete reliance on the grid.

The findings unequivocally demonstrate that incorporating renewable energy greatly lowers energy expenses and grid reliance. The hybrid system uses 40 % less energy from the grid than the standard system because of solar and biogas generation. More significantly, it sells a significant amount of its excess energy. It also emphasizes the significance of intelligent production management and energy storage. Because biogas produces more consistently and solar energy is erratic, the hybrid system maximizes grid use by restricting energy purchases to times when output is low.

Better energy independence and a business potential are indicated by the hybrid system's ability to sell its excess electricity. In addition to lowering operational costs, this improves grid stability. According to the financial research shows in Table 6, the hybrid system may drastically lower running expenses by \$17,136 year. For an energy project, the payback period of just 4.96 years is quite appealing. In general, a payback period of fewer than five years is seen as a reliable sign of economic sustainability.

Since it is significantly higher than the usual return rates for energy investments, which are typically between 10 % and 15 %, the internal rate of return (IRR) of 19.8 % is also a great indication. Fig. 14 shows that the project is not only feasible from an economic standpoint, but also has the potential to be very profitable in the long run. With a net present value (NPV) of \$136,944, the investment is guaranteed to yield substantial returns over time.

Table 6
Financial assessment of the projects.

Indicator	Value
Reduced Annual Operating Cost	\$11,803
Initial Investment	\$84,603
Return on Investment (ROI)	16.5 %
Net Present Value (NPV)	\$136,944
Internal Rate of Return (IRR)	19.8 %
Payback Period	4.96 years
Annual Savings Achieved	\$17,138

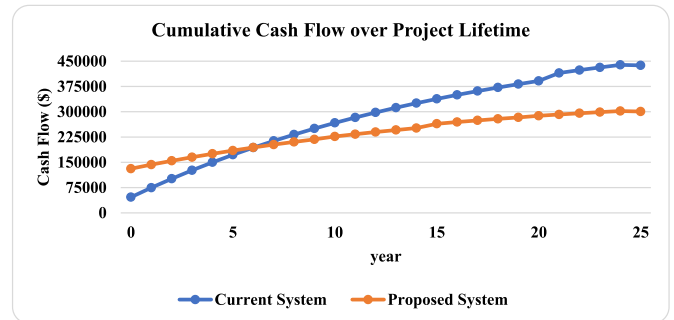


Fig. 14. Cumulative cash flow throughout the project lifetime.

4.3. Environmental cost analysis

In addition to reducing emission volumes, the hybrid system's environmental benefits can be expressed in economic terms using the carbon damage cost approach.

In the conventional configuration, where 88.9 % of energy is sourced from the grid, the annual CO₂-related environmental cost is estimated at \$4,436. By contrast, the hybrid system-which integrates 40 % PV and 25.3 % biogas-lowers grid dependency by 40 %, reducing the corresponding environmental cost to approximately \$2,309 per year.

This translates to an additional annual savings of \$2,127 in avoided carbon costs. When combined with the \$17,138 saved in operating expenses, this reinforces the economic and environmental value of adopting the hybrid solution.

The yearly emissions of pollutants produced by the hybrid system (Fig. 15b) and the traditional system (Fig. 15a) are contrasted in Fig. 15 above. When assessing the shift to a cleaner energy mix, the environmental effect is a crucial consideration. One of the primary greenhouse gases causing global warming is carbon dioxide (CO₂). The hybrid system cuts CO₂ emissions by almost half when compared to the conventional system (77,712 kg/year against 144,307 kg/year) since photovoltaics and biogas are integrated. This decline is because fewer diesel generators, a significant producer of CO₂, are being used. Additionally, there is a 44 % reduction in carbon monoxide (CO), unburned hydrocarbons, and fine particulate matter (PM), which lessens their negative effects on human health and air quality. Due to their potential to induce cardiovascular and respiratory disorders, these pollutants are especially dangerous to populations that are exposed. Nitrogen oxides (NO_x), which contribute to air pollution and the creation of smog, and sulphur dioxide (SO₂), which causes acid rain, are also decreased by 48 % and 47 %, respectively. These pollutants are lessened by the hybrid system's decreased need on fuel, which makes it more ecologically friendly.

These findings demonstrate that, in comparison to the conventional system, the hybrid system allows for a notable decrease in pollutant emissions. CO₂ emissions are cut by 48 % and other pollutants (CO, SO₂, and NO_x) are lowered by about 44 % to 48 % as a result of the



Fig. 15. Comparison of annual pollutant emissions and reduction rates between conventional and hybrid energy systems.

combination of photovoltaics and biogas (Fig. 15c). In addition to encouraging cost savings, this shift to a cleaner energy mix also greatly lessens its negative environmental effects.

Table 7 presents the comparative analysis of hybrid energy systems reveals significant variations in cost-effectiveness based on system configuration, location, and load demand. The system designed for Douala, Cameroon demonstrates a competitive LCOE of \$0.07563/kWh, outperforming similar setups in India (\$0.119/kWh), Golshan (\$0.238/kWh), and China (\$0.24/kWh), where more complex integrations and higher loads drive up costs. In contrast, the City of Chlef, Algeria achieves the lowest LCOE range (\$0.0360–\$0.0713/kWh), benefiting from efficient PV/Fuel-Cell/Grid integration and a low daily load.

Meanwhile, systems like Manoka Island, Cameroon (\$0.1981/kWh) face higher costs due to logistical challenges and multi-source complexity. Overall, Douala's system strikes an optimal balance between renewable energy integration, cost efficiency, and energy reliability.

4.4. Implementation challenges and social acceptability

Although the techno-economic analysis supports the hybrid system's feasibility, its real-world deployment depends largely on social acceptance within the campus community. Despite the scientific validity and safety of using human waste through anaerobic digestion, significant socio-cultural resistance may arise. Key challenges include cultural or

Table 7

Comparative analysis of the present HRES results with other research studies.

Refs	Hybrid System	LCOE(\$/kWh)	Load kW/d	Location
[60]	PV/Diesel/Battery	0.072	627.5	Rajshahi, Bangladesh
[61]	PV/Reformer/Fuel-Cell/grid	0.0360–0.0713	16.10 ⁶	City of Chlef, Algeria
[62]	PV/Biogas Generatorgrid	0.092	130000	Hattar, Pakistan
[51]	PV/Wind/Electrolyzer/FuelCell/Biogas/Battery	0.1981	1082.90	Manoka Island-Cameroon
[63]	PV/Diesel/Biogas/Grid	0.119	178	India
[64]	PV/Diesel/Biogas/Grid	0.238	431	Golshan
[65]	PV/Biogas/Diesel/Battery	0.24	1659.52	China
[3]	PV/Electrolyzer/Biogas	0.2094	1100	Türkiye
This study	PV/Diesel/Biogas/Battery/Grid	0.07563	580.60	Douala-Cameroon

religious taboos surrounding human excreta, concerns about hygiene and potential health risks-especially in proximity to academic spaces-and a general lack of awareness about the sanitizing and resource-converting capabilities of anaerobic digestion [40,66].

Addressing these concerns requires a deliberate and transparent stakeholder engagement strategy, which was beyond the scope of this study but is essential for future implementation. Recommended actions include educational outreach to clarify the technology, public forums to address misconceptions, and emphasizing tangible campus benefits such as energy reliability and cost savings. Support from university leadership will be vital in fostering community trust and project acceptance.

4.5. Regulatory framework and feasibility

The successful implementation of the proposed system depends not only on technical and social acceptance but also on compliance with the national regulatory framework. In Cameroon, environmental governance is primarily guided by Law No. 96/12 of August 5, 1996, which mandates the Ministry of Environment, Nature Protection and Sustainable Development (MINEPED) to oversee environmental protection and waste management [67]. While the law establishes a broad mandate for sustainable practices, it lacks detailed technical regulations specific to waste-to-energy systems using human biomass. As such, the project's regulatory viability would largely depend on obtaining approval through the Environmental and Social Impact Assessment (ESIA) process.

A comprehensive ESIA required for this type of project must demonstrate that the anaerobic digestion system effectively neutralizes pathogens, safely manages digestate by-products, and poses no health or environmental risks. Although this introduces administrative complexity, the absence of restrictive legislation, alongside national goals promoting renewable energy and improved sanitation, indicates that regulatory approval is attainable. Successfully completing the ESIA would represent a key step toward implementation.

4.6. Digestate management: a circular economy approach

A comprehensive assessment of the biogas system must include the management of its main by-product: digestate. This nutrient-rich effluent requires a sustainable strategy to prevent environmental risks and to fully leverage its resource potential. The proposed system adopts a circular economy perspective, converting digestate into a useable product through a two-stage process.

First, the raw digestate undergoes stabilization such as dewatering followed by composting to eliminate any remaining pathogens. This step ensures compliance with public health standards, including WHO guidelines for safe agricultural reuse of excreta [68,69]. Second, once treated, the digestate is repurposed as a biofertilizer, rich in N, P, and K, for application on campus landscapes and institutional gardens. This reduces dependence on synthetic fertilizers and their associated costs and environmental impacts.

Environmentally, this approach promotes nutrient recycling,

enhances soil quality, and mitigates the risks of chemical runoff. However, its success hinges on strict treatment protocols and regular monitoring to ensure the digestate is pathogen-free. As such, establishing a robust digestate management and quality control plan is essential for the system's operational phase.

4.7. Transferability and scalability of the model

An essential consideration in any strategic study is the transferability of its findings. While the system sizing in this paper is tailored to the context of Douala, the methodological framework and modular hybrid design are highly adaptable to other institutions or communities across Sub-Saharan Africa, provided appropriate adjustments are made.

Adaptability can be assessed along three key dimensions:

- **Climatic Conditions:** As a data-driven tool, HOMER Pro allows for the integration of local solar irradiance and temperature data (e.g., from NASA databases) to re-optimize system sizing. In less sunny regions, the PV component may be downsized, with increased reliance on biogas or the grid.
- **Infrastructure Context:** Grid reliability strongly influences system design. In off-grid areas, battery and diesel capacities would need to be increased to ensure continuous supply, raising both NPC and LCOE. In contrast, a robust grid enables configurations focused on cost-saving and possibly energy export.
- **Socio-Cultural Acceptability:** Human biomass use is highly context-dependent. In settings where this is culturally sensitive, the model can be adapted to other organic feedstocks (e.g., food waste, agricultural residues), or exclude biogas altogether, still delivering meaningful economic and environmental gains.

Ultimately, this study should be seen as a flexible framework rather than a fixed design, capable of guiding the development of context-specific, sustainable hybrid energy systems.

5. Conclusion

This study demonstrates the potential of an innovative hybrid energy system that integrates photovoltaic panels, a biogas generator powered by human biomass, a diesel generator for emergency backup, a battery storage unit, and a grid connection to meet the energy demands of a high-consumption facility while reducing reliance on fossil fuels. The techno-economic analysis conducted with HOMER Pro revealed substantial operational and financial benefits. Annual operating costs decreased from \$30,253.70 in the conventional configuration to \$11,803 in the hybrid system, and the levelized cost of energy (LCOE) dropped from \$0.1597/kWh to \$0.07563/kWh. The system achieves payback in 4.96 years and yields an internal rate of return (IRR) of 19.8 %. Moreover, the integration of solar and biogas technologies reduces CO₂ emissions by approximately 50 %, supporting both environmental preservation and energy transition goals. The reduced reliance on diesel further promotes efficient resource management and enhances energy

reliability.

Beyond economic performance, the analysis also quantified a marked decline in environmental externalities, underscoring the system's alignment with sustainable development principles. To further enhance system performance and autonomy, future efforts should focus on maximizing biogas production year-round by accounting for seasonal variations in biomass availability.

Equally important is addressing the social dimension of implementation. While technically and economically viable, the project's success ultimately depends on stakeholder acceptance. As this preliminary study did not include community consultation, a participatory approach involving surveys, focus groups, and educational workshops will be crucial to assess perceptions, alleviate health-related concerns, and foster a shared vision among students, staff, and decision-makers. Without this social validation, deployment risks resistance despite the project's technical merits.

Additional future work should also explore digestate management strategies, including safe treatment and reuse as biofertilizer, to fully close the resource loop. Finally, broadening the environmental scope through a full life-cycle assessment (LCA) of all system components would offer a more comprehensive understanding of long-term sustainability. This flexible strategy could serve as a replicable model for other institutions facing similar challenges in energy access and organic waste management. Comparative studies with alternative biomass sources such as food or agricultural residues could further inform system optimization and adaptability.

CRediT authorship contribution statement

Reagan Jean Jacques Molu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization, Data curation. **Inoussah Moungnoutou Mfetoum:** Writing – original draft, Writing – review & editing. **Paul Lionnel Ella Nkoua:** Writing – original draft. **Serge Raoul Dzone Naoussi:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Y.F. Nassar, H.J. El-Khozondar, M. Elnaggar, F.F. El-batta, R.J. El-Khozondar, et al. S.Y. Alsadi, Renewable energy potential in the State of Palestine: proposals for sustainability, *Renew. Energy Focus* 49 (juin 2024) 100576, <https://doi.org/10.1016/j.ref.2024.100576>.
- [2] Md Islam, H. Akter, H. Howlader, et al. T. Senjyu, Optimal sizing and techno-economic analysis of grid-independent hybrid energy system for sustained rural electrification in developing countries: a case study in Bangladesh, *Energies* 15 (17) (sept. 2022) 6381, <https://doi.org/10.3390/en15176381>.
- [3] B. Hüner, Feasibility and environmental analysis of biogas-based hybrid energy system using HOMER pro software: a case study for Hatay, *Energy Convers. Manag.* 326 (févr. 2025) 119480, <https://doi.org/10.1016/j.enconman.2025.119480>.
- [4] J. Yi, S. Dai, L. Li, et al. J. Cheng, How does digital economy development affect renewable energy innovation? *Renew. Sustain. Energy Rev.* 192 (mars 2024) 114221 <https://doi.org/10.1016/j.rser.2023.114221>.
- [5] R.J.J. Molu, S. Raoul Dzone Naoussi, P. Wira, W.F. Mbasso, S.T. Kenfack, et al. S. Kamel, Optimization-based energy management system for grid-connected photovoltaic/battery microgrids under uncertainty, *Case Stud. Chem. Environ. Eng.* 8 (déc. 2023) 100464, <https://doi.org/10.1016/j.csee.2023.100464>.
- [6] R.J.J. Molu, et al., A techno-economic perspective on efficient hybrid renewable energy solutions in Douala, Cameroon's grid-connected systems, *Sci. Rep.* 14 (1) (juin 2024) 13590, <https://doi.org/10.1038/s41598-024-64427-4>.
- [7] F.M. Wulfran, D.N.S. Raoul, M.R.J. Jacques, K.T. Saatong, et al. S. Kamel, A case study of a neighborhood in Douala examining the technical feasibility of a hybrid renewable energy system connected to the grid for energy and oxygen production in the fight against respiratory diseases, *Sol. Compass* 8 (déc. 2023) 100062, <https://doi.org/10.1016/j.solcom.2023.100062>.
- [8] Z. Ul Islam, et al., Optimizing hybrid renewable energy based automated railway level crossing in Bangladesh: techno-economic, emission and sensitivity analysis, *Energy Convers. Manag.* X 24 (oct. 2024) 100744, <https://doi.org/10.1016/j.ecmx.2024.100744>.
- [9] N. Ul Rehman, et al. H. Farzaneh, Techno-economic analysis of a hybrid heat recovery-renewable energy system for enhancing power reliability in cement factories in Pakistan, *Energy Convers. Manag.* X 20 (oct. 2023) 100492, <https://doi.org/10.1016/j.ecmx.2023.100492>.
- [10] A. Sadeghi, A. Maleki, et al. S. Haghghat, Techno-economic analysis and optimization of a hybrid solar-wind-biomass-battery framework for the electrification of a remote area: a case study, *Energy Convers. Manag.* X 24 (oct. 2024) 100732, <https://doi.org/10.1016/j.ecmx.2024.100732>.
- [11] M. Ashrafur Islam, et al., Assessing the feasibility and quality performance of a renewable Energy-Based hybrid microgrid for electrification of remote communities, *Energy Convers. Manag.* X 23 (juill. 2024) 100674, <https://doi.org/10.1016/j.ecmx.2024.100674>.
- [12] U. Singh, et al. M. Rizwan, A feasibility study and cost-benefit analysis of an off-grid hybrid system for a remote area electrification, *Future Energy* 2 (2) (2023) 2.
- [13] T. Ali, M. Reaz Sunny, K. Aghaloo, et al. K. Wang, Planning off-grid hybrid energy system using techno-economic optimization and wins in league theory-based multicriteria decision-making method in the wetland areas of developing countries, *Energy Convers. Manag.* 313 (août 2024) 118587, <https://doi.org/10.1016/j.enconman.2024.118587>.
- [14] Md F.H. Mojumder, et al., Techno-economic and environmental analysis of hybrid energy systems for remote areas: a sustainable case study in Bangladesh, *Energy Convers. Manag.* X 23 (juill. 2024) 100664, <https://doi.org/10.1016/j.ecmx.2024.100664>.
- [15] B.A. Medjo Nouadje, P. Tiam Kapen, V. Chegnimonhan, et al. R. Tchinda, Techno-economic analysis of a grid/fuel cell/PV/electrolyzer system for hydrogen and electricity production in the countries of African and Malagasy council for higher education (CAMES), *Energy Strategy Rev.* 53 (mai 2024) 101392, <https://doi.org/10.1016/j.esr.2024.101392>.
- [16] A. El-Maaroufi, M. Daoudi, et al. R. Ahl Laamara, Techno-economic analysis of a PV/WT/biomass off-grid hybrid power system for rural electrification in northern Morocco using HOMER, *Renew. Energy* 231 (sept. 2024) 120904, <https://doi.org/10.1016/j.renene.2024.120904>.
- [17] S. Yadav, P. Kumar, et al. A. Kumar, Techno-economic assessment of hybrid renewable energy system with multi energy storage system using HOMER, *Energy* 297 (juin 2024) 131231, <https://doi.org/10.1016/j.energy.2024.131231>.
- [18] S.S. Chamazkoti, A. Hajinezhad, et al. S.F. Moosavian, Feasibility study of a flexible hybrid energy model with power-hydrogen production in wastewater treatment plant, *Energy Rep.* 11 (juin 2024) 3189–3201, <https://doi.org/10.1016/j.egy.2024.02.062>.
- [19] P. Kumar, N. Pal, et al. H. Sharma, Optimization and techno-economic analysis of a solar photo-voltaic/biomass/diesel/battery hybrid off-grid power generation system for rural remote electrification in eastern India, *Energy* 247 (mai 2022) 123560, <https://doi.org/10.1016/j.energy.2022.123560>.
- [20] H.A. El-Sattar, S. Kamel, H.M. Sultan, H.M. Zawbaa, et al. F. Jurado, Optimal design of photovoltaic, biomass, fuel cell, hydrogen tank units and electrolyzer hybrid system for a remote area in Egypt, *Energy Rep.* 8 (nov. 2022) 9506–9527, <https://doi.org/10.1016/j.egy.2022.07.060>.
- [21] M. Nyasapoh, et al., Navigating renewable energy transition challenges for a sustainable energy future in Ghana, *Sol. Energy Sustain. Dev. J.* 14 (1) (mars 2025) 237–257, <https://doi.org/10.51646/jesd.v14i1.479>.
- [22] M. Abuhelwa, M. Elnaggar, W.A. Salah, Y.F. Nassar, et al. M.J.K. Bashir, Exploring the prevalence of renewable energy practices and awareness levels in Palestine, *Energy Sci. Eng.* 13 (3) (mars 2025) 1292–1305, <https://doi.org/10.1002/ese3.2070>.
- [23] M.A. Nyasapoh, et al., Integrated assessment of nuclear-renewable hybrid energy systems: a pathway to sustainable and resilient industrial electrification in Ghana, *Afr. J. Appl. Res.* 11 (2) (mars 2025) 22–46, <https://doi.org/10.26437/ajar.v11i2.969>.
- [24] H.J. El-Khozondar, F. El-batta, R.J. El-Khozondar, Y. Nassar, M. Alramlawi, et al. S. Alsadi, Standalone hybrid PV/wind/diesel-electric generator system for a COVID-19 quarantine center, *Environ. Prog. Sustain. Energy* 42 (3) (mai 2023) e14049, <https://doi.org/10.1002/ep.14049>.
- [25] A. Aqila, Y. Nassar, H. El-Khozondar, et al. S. Suliman, Design of hybrid renewable energy system (PV/Wind/Battery) under real climatic and operational conditions to meet full load of the residential sector: a case study of a house in Samno village-southern region of Libya », *Wadi Alshatti Univ. J. Pure Appl. Sci.*, p. 168–181, avr (2025), <https://doi.org/10.63318/waujpasv31i.23>.
- [26] H. Awad, Y.F. Nassar, A. Hafez, M.K. Sherbiny, et al. F.M. Ali, Optimal design and economic feasibility of rooftop photovoltaic energy system for Assuit University, Egypt, *Ain Shams Eng. J.* 13 (3) (mai 2022) 101599, <https://doi.org/10.1016/j.asej.2021.09.026>.
- [27] A. Alsharif, E. Almasbout, A.A. Ahmed, M. Khaleel, Y.F. Nassar, et al. H.J. El-Khozad, Optimal sizing of hybrid renewable system for residential appliances, in: 2024 IEEE 4th International Maghreb Meeting of the Conference on Sciences and Techniques of Automatic Control and Computer Engineering (MI-STA), Tripoli, Libya, IEEE, mai 2024, pp. 290–295, <https://doi.org/10.1109/MI-STA61267.2024.10599729>.
- [28] E. Salim, A. Abubaker, B. Ahmed, et al. Y. Nassar, A brief overview of hybrid renewable energy systems and analysis of integration of isolated hybrid PV solar

- system with pumped hydropower storage for Brack city - Libya », Wadi Alshatti Univ. J. Pure Appl. Sci. (2025) https://doi.org/10.63318/waujpasv3i1_22, p. 152-167, avr.
- [29] Y.F. Nassar, et al., Design of reliable standalone utility-scale pumped hydroelectric storage powered by PV/Wind hybrid renewable system, *Energy Convers. Manag.* 322 (déc. 2024) 119173, <https://doi.org/10.1016/j.enconman.2024.119173>.
- [30] Y.F. Nassar, H.J. El-Khozondar, et al.M.A. Fakher, The role of hybrid renewable energy systems in covering power shortages in public electricity grid: an economic, environmental and technical optimization analysis, *J. Energy Storage* 108 (févr. 2025) 115224, <https://doi.org/10.1016/j.est.2024.115224>.
- [31] College of Renewable Energy, Libya Tajoura, Renewable energy homes generating as a sustainable solution to meet Libya's household energy needs, *Int. Sci. Technol. J.* 35 (1) (oct. 2024) 1–12, <https://doi.org/10.62341/licase2039>.
- [32] A.F.M. Ali, E.M.H. Karraam, Y.F. Nassar, et al.A.A. Hafez, « reliable and economic isolated renewable hybrid power system with pumped hydropower storage, in: 2021 22nd International Middle East Power Systems Conference (MEPCON), Assiut, Egypt, IEEE, déc. 2021, pp. 515–520, <https://doi.org/10.1109/MEPCON50283.2021.9686233>.
- [33] M.M. Al-Maghalseh, A.K. Abutemeah, et M. Iyadyyeh, « Modeling a hybrid system for electrical generation and wastewater treatment using photovoltaic and fuel cells, *Desalination Water Treat.* 100 (déc. 2017) 250–257, <https://doi.org/10.5004/dwt.2017.21315>.
- [34] Y.F. Nassar, et al., Dynamic analysis and sizing optimization of a pumped hydroelectric storage-integrated hybrid PV/Wind system: a case study, *Energy Convers. Manag.* 229 (févr. 2021) 113744, <https://doi.org/10.1016/j.enconman.2020.113744>.
- [35] M.A. Eteriki, W.A. El-Osta, Y.F. Nassar, et al.H.J.E. Khozondar, Effect of implementation of energy efficiency in residential sector in Libya, in: 2023 8th International Engineering Conference on Renewable Energy & Sustainability (ieCRES), Gaza, Palestine, State of, IEEE, mai 2023, pp. 1–6, <https://doi.org/10.1109/ieCRES57315.2023.10209521>.
- [36] M. Abdunnabi, N. Etiab, Y.F. Nassar, H.J. El-Khozondar, et al.R. Khargotra, Energy savings strategy for the residential sector in Libya and its impacts on the global environment and the nation economy, *Adv. Build. Energy Res.* 17 (4) (juill. 2023) 379–411, <https://doi.org/10.1080/17512549.2023.2209094>.
- [37] C.A. Moraes, L. de L.C. dos Santos, A.C.L. de Oliveira, D.F. Botelho, J. Moltó Berenguer, et al.N. dos S. Renato, « Biogas-based electricity production from landfills in places of irregular disposal: overview for the southeast region of Brazil, *Energy* 290 (mars 2024) 130161, <https://doi.org/10.1016/j.energy.2023.130161>.
- [38] H.M.A. Shahzad, F. Almomani, A. Shahzad, K.A. Mahmoud, et K. Rasool, Challenges and opportunities in biogas conversion to microbial protein: a pathway for sustainable resource recovery from organic waste, *Process Saf. Environ. Prot.* 185 (mai 2024) 644–659, <https://doi.org/10.1016/j.psep.2024.03.055>.
- [39] M. Imran, Z. Rashid, M. Amjad, S.K. Baloch, et al.A. Mustafa, Analysis and implementation of a solar/biogas hybrid system for centralized generation at Qadirpur Ran, *Energy Rep.* 8 (nov. 2022) 13484–13493, <https://doi.org/10.1016/j.egy.2022.10.017>.
- [40] M. Kharrich, O. Mohammed, et al.M. Akherraz, Design of hybrid microgrid PV/Wind/Diesel/Battery system: case study for Rabat and Baghdad, *EAI Endorsed Trans. Energy Web* 0 (0) (juill. 2018) 162692, <https://doi.org/10.4108/eai.13-7-2018.162692>.
- [41] M. Elnaggar, H.J. El-Khozondar, W.A. Salah, Y.F. Nassar, et al.M.J.K. Bashir, Assessing the Techno-enviro-economic viability of wind farms to address electricity shortages and Foster sustainability in Palestine, *Results Eng.* 24 (déc. 2024) 103111, <https://doi.org/10.1016/j.rineng.2024.103111>.
- [42] A. Ajlan, C.W. Tan, et al.A.M. Abdilahi, Assessment of environmental and economic perspectives for renewable-based hybrid power system in Yemen, *Renew. Sustain. Energy Rev.* 75 (août 2017) 559–570, <https://doi.org/10.1016/j.rser.2016.11.024>.
- [43] C.I. Aragón-Briceño, O. Grasham, A.B. Ross, V. Dupont, et al.M.A. Camargo-Valero, Hydrothermal carbonization of sewage digestate at wastewater treatment works: influence of solid loading on characteristics of hydrochar, process water and plant energetics, *Renew. Energy* 157 (sept. 2020) 959–973, <https://doi.org/10.1016/j.renene.2020.05.021>.
- [44] D. Nayeri, P. Mohammadi, P. Bashardoust, et al.N. Eshtiahi, A comprehensive review on the recent development of anaerobic sludge digestions: performance, mechanism, operational factors, and future challenges, *Results Eng.* 22 (juin 2024) 102292, <https://doi.org/10.1016/j.rineng.2024.102292>.
- [45] E. Lee, et al., Biogas production from high solids anaerobic co-digestion of food waste, yard waste and waste activated sludge, *Waste Manag.* 95 (juill. 2019) 432–439, <https://doi.org/10.1016/j.wasman.2019.06.033>.
- [46] L. Adelard, T.G. Poulsen, et al.V. Rakotoniaina, « Biogas and methane yield in response to co- and separate digestion of biomass wastes, *Waste Manag. Res. J. Sustain. Circ. Econ.* 33 (1) (janv. 2015) 55–62, <https://doi.org/10.1177/0734242X14559406>.
- [47] C.T. Mutate, A.J. Kanjanda, et al.G. Mehta, Small-scale electricity generation from biogas in third world countries, in: R. Sharma, R. Kannojiya, N. Garg, et al.S. Gautam (Eds.), *Advances in Engineering Design*, Springer Nature, Singapore, 2023, pp. 449–460, https://doi.org/10.1007/978-981-99-3033-3_38.
- [48] S. Mukhtar, S. Muhammad, H.A. Alyousef, W. Khan, R. Shah, et al.S.A. El-Tantawy, Enviro-economic and optimal hybrid energy system: photovoltaic-biogas-hydro-battery system in rural areas of Pakistan, *Heliyon* 10 (16) (août 2024) e35182, <https://doi.org/10.1016/j.heliyon.2024.e35182>.
- [49] C. Yan, Y. Zou, Z. Wu, et al., Maleki, « Effect of various design configurations and operating conditions for optimization of a wind/solar/hydrogen/fuel cell hybrid microgrid system by a bio-inspired algorithm, *Int. J. Hydrogen Energy* 60 (mars 2024) 378–391, <https://doi.org/10.1016/j.ijhydene.2024.02.004>.
- [50] B. Hüner, Mathematical modeling of an integrated photovoltaic-assisted PEM water electrolyzer system for hydrogen production, *Int. J. Hydrogen Energy* 79 (août 2024) 594–608, <https://doi.org/10.1016/j.ijhydene.2024.07.041>.
- [51] R.J. Jacques Molu, et al., Optimizing technical and economic aspects of off-grid hybrid renewable systems: a case study of Manoka Island, Cameroon, *IEEE Access* 11 (2023) 130909–130930, <https://doi.org/10.1109/ACCESS.2023.3332693>.
- [52] M. Kaur, S. Dhundhara, Y.P. Verma, et al.S. Chauhan, Techno-economic analysis of photovoltaic-biomass-based microgrid system for reliable rural electrification, *Int. Trans. Electr. Energy Syst.* 30 (5) (mai 2020), <https://doi.org/10.1002/2050-7038.12347>.
- [53] O.D.T. Odou, R. Bhandari, et al.R. Adamou, Hybrid off-grid renewable power system for sustainable rural electrification in Benin, *Renew. Energy* 145 (janv. 2020) 1266–1279, <https://doi.org/10.1016/j.renene.2019.06.032>.
- [54] S. Rajanna, et al.R.P. Saini, Development of optimal integrated renewable energy model with battery storage for a remote Indian area, *Energy* 111 (sept. 2016) 803–817, <https://doi.org/10.1016/j.energy.2016.06.005>.
- [55] S. Mokhtara, B. Negrou, A. Boufferrouk, Y. Yao, N. Settou, et al.M. Ramadan, Integrated supply-demand energy management for optimal design of off-grid hybrid renewable energy systems for residential electrification in arid climates, *Energy Convers. Manag.* 221 (oct. 2020) 113192, <https://doi.org/10.1016/j.enconman.2020.113192>.
- [56] A. Abodwair, M. Guneser, M. Khaleel, Y. Nassar, H. El-Khozondar, et al., Elbaz, « feasibility assessment of hybrid renewable energy based EV charging station in Libya, *Sol. Energy Sustain. Dev. J.* 13 (2) (nov. 2024) 311–349, <https://doi.org/10.51646/jesd.v13i2.292>.
- [57] S. Mohammed, Y. Nassar, H. Algassie, A. Mahammed, H. El-Khozondar, et al. A. Alatrash, Exploring promised sites for establishing hydropower energy storage (PHES) stations in Libya by using the geographic information systems (GIS), *Wadi Alshatti Univ. J. Pure Appl. Sci.* (mars 2025) 85–94, https://doi.org/10.63318/waujpasv3i1_13.
- [58] Y. Nassar, et al., Towards green economy: case of electricity generation sector in Libya, *Sol. Energy Sustain. Dev. J.* 14 (1) (2025) 334–360, <https://doi.org/10.51646/jesd.v14i1.549>, avr.
- [59] Y.F. Nassar, et al., Design of an isolated renewable hybrid energy system: a case study, *Mater. Renew. Sustain. Energy* 11 (3) (2022) 225–240, <https://doi.org/10.1007/s40243-022-00216-1>, déc.
- [60] B.K. Das, M.A. Alotaibi, P. Das, M.S. Islam, S.K. Das, et al.M.A. Hossain, Feasibility and techno-economic analysis of stand-alone and grid-connected PV/Wind/Diesel/Batt hybrid energy system: a case study, *Energy Strategy Rev.* 37 (sept. 2021) 100673, <https://doi.org/10.1016/j.esr.2021.100673>.
- [61] M. Dekkiche, T. Tahri, et al.M. Denai, Techno-economic comparative study of grid-connected PV/reformer/FC hybrid systems with distinct solar tracking systems, *Energy Convers. Manag.* X 18 (avr. 2023) 100360, <https://doi.org/10.1016/j.ecmx.2023.100360>.
- [62] A. Zahid, M.K. Shahzad, S.R. Jamil, et N. Iqbal, Futuristic feasibility analysis and modelling of a solar-biomass on-grid hybrid system for Hattar Industrial Estate Phase (VII), Pakistan », *Clean, Energy Syst.* 4 (avr. 2023) 100053, <https://doi.org/10.1016/j.cles.2023.100053>.
- [63] R. Rajbongshi, D. Borgohain, et al.S. Mahapatra, Optimization of PV-biomass-diesel and grid base hybrid energy systems for rural electrification by using HOMER, *Energy* 126 (mai 2017) 461–474, <https://doi.org/10.1016/j.energy.2017.03.056>.
- [64] A. Kasaeian, P. Rahdan, M.A.V. Rad, et al.W.-M. Yan, Optimal design and technical analysis of a grid-connected hybrid photovoltaic/diesel/biogas under different economic conditions: a case study, *Energy Convers. Manag.* 198 (oct. 2019) 111810, <https://doi.org/10.1016/j.enconman.2019.111810>.
- [65] C. Li, L. Zhang, F. Qiu, et al.R. Fu, Optimization and enviro-economic assessment of hybrid sustainable energy systems: the case study of a photovoltaic/biogas/diesel/battery system in Xuzhou, China, *Energy Strategy Rev.* 41 (mai 2022) 100852, <https://doi.org/10.1016/j.esr.2022.100852>.
- [66] F. Kemausuor, S. Bolwig, et al.S. Miller, Modelling the socio-economic impacts of modern bioenergy in rural communities in Ghana, *Sustain. Energy Technol. Assess.* 14 (avr. 2016) 9–20, <https://doi.org/10.1016/j.seta.2016.01.007>.
- [67] Loi n° 96-12 portant loi-cadre relative à la gestion de l'environnement. [FAOLEX ». Consulté le: 10 juin 2025 [En ligne]. Disponible sur: <https://www.fao.org/faolex/results/details/en/c/LEX-FAOC019578/>.
- [68] Guidelines for the safe use of wastewater, excreta and greywater Volume 4 Excreta and greywater use in agriculture ». Consulté le: 10 juin 2025 [En ligne]. Disponible sur: <https://www.who.int/publications/i/item/9241546859>.
- [69] R. Arthur, M.F. Baidoo, et al.E. Antwi, Biogas as a potential renewable energy source: a Ghanaian case study, *Renew. Energy* 36 (5) (mai 2011) 1510–1516, <https://doi.org/10.1016/j.renene.2010.11.012>.